a

b

c

SCS

GBR

CARIB

top: TSAR model response

bottom: residual R = TSAR − PersA

ΔDHW over last 4 weeks (°C-weeks)
← falling | rising →

Current DHW (°C-weeks)

TSAR (counterfactual)

R = TSAR − PersA

d

e

f

Counterfactual model response of the frozen TSAR, not a causal effect.
Grey = outside real-data support; thin contours = support density.